```
 1  using System;
 2  using System.Drawing;
 3  using System.Windows.Forms;
 4
 5  namespace RogueOfTheDarkAges
 6  {
 7      public class MainForm : Form
 8      {
 9          private Random random;
10          private PlayerClass player;
11          private Enemy currentEnemy;
12          private int currentLevel;
13          private bool bossBattle;
14          private bool waitingForUpgrade;
15          private bool gameOver;
16
17          private string[] environments;
18          private string[] enemyNames;
19
20          private Label titleLabel;
21          private Label setupHeaderLabel;
22          private Label playerHeaderLabel;
23          private Label enemyHeaderLabel;
24          private Label actionsHeaderLabel;
25          private Label upgradeHeaderLabel;
26          private Label logHeaderLabel;
27          private Label statusLabel;
28
29          private Label classSelectLabel;
30          private Label levelLabel;
31          private Label classLabel;
32          private Label playerHealthLabel;
33          private Label weaponLabel;
34          private Label bonusDamageLabel;
35          private Label potionLabel;
36          private Label enemyNameLabel;
37          private Label enemyHealthLabel;
38          private Label environmentLabel;
39
40          private ProgressBar playerHealthBar;
41          private ProgressBar enemyHealthBar;
42
43          private ComboBox classSelector;
44          private Button startButton;
45          private Button restartButton;
46          private Button attackButton;
47          private Button blockButton;
48          private Button healButton;
49          private Button upgradeHealthButton;
```

```
50          private Button upgradeDamageButton;
51          private Button upgradePotionButton;
52
53          private RichTextBox logBox;
54
55          public MainForm()
56          {
57              random = new Random();
58
59              environments = new string[]
60              {
61                  "a fog-covered swamp where the earth squelches beneath
                      your boots",
62                  "the ruins of an old watchtower scattered with broken
                      stone",
63                  "a pine forest clearing lit by a pale morning sun",
64                  "a muddy battlefield abandoned after a brutal clash",
65                  "a narrow mountain pass with cold wind cutting through
                      the rocks",
66                  "an overgrown monastery courtyard filled with creeping
                      ivy",
67                  "a torchlit castle hallway smelling of smoke and iron",
68                  "a river crossing where the current crashes against worn
                      pillars",
69                  "a village square left eerily silent after nightfall",
70                  "a windswept cliffside where gulls cry over the sea"
71              };
72
73              enemyNames = new string[]
74              {
75                  "Brigand",
76                  "Mercenary",
77                  "Raider",
78                  "Footman",
79                  "Pirate",
80                  "Hunter",
81                  "Knight",
82                  "Marauder",
83                  "Bandit Captain",
84                  "Grave Robber",
85                  "Soldier",
86                  "Guard",
87                  "Scout",
88                  "Warrior"
89              };
90
91              BuildInterface();
92              ResetToMenu();
93          }
```

```
 94
 95          private void BuildInterface()
 96          {
 97              Text = "Rogue of the Dark Ages";
 98              StartPosition = FormStartPosition.CenterScreen;
 99              ClientSize = new Size(1120, 820);
100              FormBorderStyle = FormBorderStyle.FixedSingle;
101              MaximizeBox = false;
102              BackColor = Color.FromArgb(20, 22, 28);
103              ForeColor = Color.White;
104              Font = new Font("Segoe UI", 10F, FontStyle.Regular);
105
106              titleLabel = new Label();
107              titleLabel.Text = "ROGUE OF THE DARK AGES";
108              titleLabel.Font = new Font("Segoe UI", 22F, FontStyle.Bold);
109              titleLabel.ForeColor = Color.Gold;
110              titleLabel.AutoSize = true;
111              titleLabel.Location = new Point(24, 18);
112              Controls.Add(titleLabel);
113
114              statusLabel = new Label();
115              statusLabel.Font = new Font("Segoe UI", 10F, FontStyle.Bold);
116              statusLabel.ForeColor = Color.LightSkyBlue;
117              statusLabel.AutoSize = false;
118              statusLabel.Location = new Point(28, 58);
119              statusLabel.Size = new Size(1040, 24);
120              Controls.Add(statusLabel);
121
122              setupHeaderLabel = CreateSectionLabel("Game Setup", 28, 95);
123              Controls.Add(setupHeaderLabel);
124
125              classSelectLabel = new Label();
126              classSelectLabel.Text = "Choose Class:";
127              classSelectLabel.Location = new Point(30, 128);
128              classSelectLabel.Size = new Size(100, 24);
129              classSelectLabel.ForeColor = Color.WhiteSmoke;
130              Controls.Add(classSelectLabel);
131
132              classSelector = new ComboBox();
133              classSelector.DropDownStyle = ComboBoxStyle.DropDownList;
134              classSelector.Location = new Point(135, 125);
135              classSelector.Size = new Size(180, 28);
136              classSelector.Items.Add("Knight");
137              classSelector.Items.Add("Vanguard");
138              classSelector.Items.Add("Hunter");
139              classSelector.Items.Add("Rogue");
140              classSelector.Items.Add("Mage");
141              classSelector.SelectedIndex = 0;
142              Controls.Add(classSelector);
```

```
143
144             startButton = CreateButton("Start Game", 330, 123, 120, 34);
145             startButton.Click += StartButton_Click;
146             Controls.Add(startButton);
147
148             restartButton = CreateButton("Restart", 460, 123, 100, 34);
149             restartButton.Click += RestartButton_Click;
150             Controls.Add(restartButton);
151
152             playerHeaderLabel = CreateSectionLabel("Player", 28, 180);
153             Controls.Add(playerHeaderLabel);
154
155             levelLabel = CreateInfoLabel(30, 214, 460);
156             Controls.Add(levelLabel);
157
158             classLabel = CreateInfoLabel(30, 244, 460);
159             Controls.Add(classLabel);
160
161             playerHealthLabel = CreateInfoLabel(30, 274, 460);
162             Controls.Add(playerHealthLabel);
163
164             playerHealthBar = new ProgressBar();
165             playerHealthBar.Location = new Point(30, 304);
166             playerHealthBar.Size = new Size(300, 18);
167             playerHealthBar.Style = ProgressBarStyle.Continuous;
168             Controls.Add(playerHealthBar);
169
170             weaponLabel = CreateInfoLabel(30, 334, 460);
171             Controls.Add(weaponLabel);
172
173             bonusDamageLabel = CreateInfoLabel(30, 364, 460);
174             Controls.Add(bonusDamageLabel);
175
176             potionLabel = CreateInfoLabel(30, 394, 460);
177             Controls.Add(potionLabel);
178
179             enemyHeaderLabel = CreateSectionLabel("Enemy", 540, 180);
180             Controls.Add(enemyHeaderLabel);
181
182             enemyNameLabel = CreateInfoLabel(542, 214, 540);
183             Controls.Add(enemyNameLabel);
184
185             enemyHealthLabel = CreateInfoLabel(542, 244, 540);
186             Controls.Add(enemyHealthLabel);
187
188             enemyHealthBar = new ProgressBar();
189             enemyHealthBar.Location = new Point(542, 274);
190             enemyHealthBar.Size = new Size(300, 18);
191             enemyHealthBar.Style = ProgressBarStyle.Continuous;
```

```
192             Controls.Add(enemyHealthBar);
193
194             environmentLabel = new Label();
195             environmentLabel.Location = new Point(542, 304);
196             environmentLabel.Size = new Size(540, 90);
197             environmentLabel.ForeColor = Color.LightCyan;
198             environmentLabel.AutoSize = false;
199             Controls.Add(environmentLabel);
200
201             actionsHeaderLabel = CreateSectionLabel("Actions", 28, 450);
202             Controls.Add(actionsHeaderLabel);
203
204             attackButton = CreateButton("Attack", 30, 485, 130, 42);
205             attackButton.Click += AttackButton_Click;
206             Controls.Add(attackButton);
207
208             blockButton = CreateButton("Block", 180, 485, 130, 42);
209             blockButton.Click += BlockButton_Click;
210             Controls.Add(blockButton);
211
212             healButton = CreateButton("Heal", 330, 485, 130, 42);
213             healButton.Click += HealButton_Click;
214             Controls.Add(healButton);
215
216             upgradeHeaderLabel = CreateSectionLabel("Upgrade Choice", 28,
                  560);
217             Controls.Add(upgradeHeaderLabel);
218
219             upgradeHealthButton = CreateButton("+10 Max Health", 30, 595,
                  150, 42);
220             upgradeHealthButton.Click += UpgradeHealthButton_Click;
221             Controls.Add(upgradeHealthButton);
222
223             upgradeDamageButton = CreateButton("+2 Damage", 190, 595,
                  130, 42);
224             upgradeDamageButton.Click += UpgradeDamageButton_Click;
225             Controls.Add(upgradeDamageButton);
226
227             upgradePotionButton = CreateButton("+1 Potion", 330, 595,
                  130, 42);
228             upgradePotionButton.Click += UpgradePotionButton_Click;
229             Controls.Add(upgradePotionButton);
230
231             logHeaderLabel = CreateSectionLabel("Battle Log", 560, 430);
232             Controls.Add(logHeaderLabel);
233
234             logBox = new RichTextBox();
235             logBox.Location = new Point(560, 470);
236             logBox.Size = new Size(520, 315);
```

```
237             logBox.ReadOnly = true;
238             logBox.Font = new Font("Consolas", 10F, FontStyle.Regular);
239             logBox.BackColor = Color.FromArgb(12, 14, 18);
240             logBox.ForeColor = Color.WhiteSmoke;
241             logBox.BorderStyle = BorderStyle.FixedSingle;
242             logBox.ScrollBars = RichTextBoxScrollBars.Vertical;
243             logBox.DetectUrls = false;
244             Controls.Add(logBox);
245         }
246 
247         private Label CreateSectionLabel(string text, int x, int y)
248         {
249             Label label = new Label();
250             label.Text = text;
251             label.Font = new Font("Segoe UI", 11F, FontStyle.Bold);
252             label.ForeColor = Color.Goldenrod;
253             label.Location = new Point(x, y);
254             label.Size = new Size(280, 24);
255             return label;
256         }
257 
258         private Label CreateInfoLabel(int x, int y, int width)
259         {
260             Label label = new Label();
261             label.Location = new Point(x, y);
262             label.Size = new Size(width, 24);
263             label.ForeColor = Color.WhiteSmoke;
264             return label;
265         }
266 
267         private Button CreateButton(string text, int x, int y, int width,
              int height)
268         {
269             Button button = new Button();
270             button.Text = text;
271             button.Location = new Point(x, y);
272             button.Size = new Size(width, height);
273             button.FlatStyle = FlatStyle.Flat;
274             button.FlatAppearance.BorderSize = 1;
275             button.FlatAppearance.BorderColor = Color.FromArgb(80, 90,
                  110);
276             button.BackColor = Color.FromArgb(38, 43, 54);
277             button.ForeColor = Color.White;
278             button.Cursor = Cursors.Hand;
279             return button;
280         }
281 
282         private void ResetToMenu()
283         {
```

```
284             player = null;
285             currentEnemy = null;
286             currentLevel = 0;
287             bossBattle = false;
288             waitingForUpgrade = false;
289             gameOver = false;
290 
291             levelLabel.Text = "Level: -";
292             classLabel.Text = "Class: -";
293             playerHealthLabel.Text = "Health: -";
294             weaponLabel.Text = "Weapon: -";
295             bonusDamageLabel.Text = "Bonus Damage: -";
296             potionLabel.Text = "Potions: -";
297 
298             enemyNameLabel.Text = "Enemy: -";
299             enemyHealthLabel.Text = "Enemy Health: -";
300             environmentLabel.Text = "Environment: Choose a class and
                  start a new run.";
301 
302             playerHealthBar.Value = 0;
303             enemyHealthBar.Value = 0;
304 
305             logBox.Clear();
306             AddHeaderLog("Welcome to Rogue of the Dark Ages.");
307             AddInfoLog("Block reduces incoming damage to 25% and has a
                  25% chance to counter.");
308             AddInfoLog("Choose a class, then press Start Game.");
309 
310             statusLabel.Text = "Ready to begin.";
311             SetBattleButtons(false);
312             SetUpgradeButtons(false);
313 
314             startButton.Enabled = true;
315             classSelector.Enabled = true;
316         }
317 
318         private void StartButton_Click(object sender, EventArgs e)
319         {
320             player = CreateSelectedPlayerClass();
321             player.Potions = 4;
322             player.BonusDamage = 0;
323             currentLevel = 1;
324             bossBattle = false;
325             waitingForUpgrade = false;
326             gameOver = false;
327 
328             logBox.Clear();
329             AddPlayerLog("You chose the " + player.ClassName + ".");
330             AddInfoLog("Starting weapon: " + player.Weapon.Name);
```

```
331             AddInfoLog(player.GetClassDescription());
332             AddInfoLog("You start with 4 potions.");
333 
334             startButton.Enabled = false;
335             classSelector.Enabled = false;
336 
337             StartCurrentLevel();
338         }
339 
340         private void RestartButton_Click(object sender, EventArgs e)
341         {
342             ResetToMenu();
343         }
344 
345         private PlayerClass CreateSelectedPlayerClass()
346         {
347             string selectedClass = classSelector.SelectedItem.ToString();
348 
349             if (selectedClass == "Knight")
350             {
351                 return new Knight();
352             }
353             else if (selectedClass == "Vanguard")
354             {
355                 return new Vanguard();
356             }
357             else if (selectedClass == "Hunter")
358             {
359                 return new Hunter();
360             }
361             else if (selectedClass == "Rogue")
362             {
363                 return new Rogue();
364             }
365             else
366             {
367                 return new Mage();
368             }
369         }
370 
371         private void StartCurrentLevel()
372         {
373             waitingForUpgrade = false;
374             SetUpgradeButtons(false);
375             SetBattleButtons(true);
376 
377             player.RestoreHealthToFull();
378 
379             if (currentLevel <= 6)
```

```
380             {
381                 bossBattle = false;
382                 currentEnemy = CreateEnemy(currentLevel);
383                 string place = environments[random.Next
                      (environments.Length)];
384 
385                 AddLog("");
386                 AddHeaderLog("=== Level " + currentLevel + " ===");
387                 AddInfoLog("You arrive in " + place + ".");
388                 AddEnemyLog("A " + currentEnemy.Name + " " +
                      currentEnemy.EnemyClassName + " appears with a " +
                      currentEnemy.Weapon.Name + ".");
389                 environmentLabel.Text = "Environment: " + place;
390                 statusLabel.Text = "Choose an action: Attack, Block, or
                      Heal.";
391             }
392             else
393             {
394                 bossBattle = true;
395                 currentEnemy = new Enemy("Ancient Dragon", "Boss", 95,
                      new Weapon(WeaponType.IgnitedClaws, "Ignited Claws",
                      10, 15));
396                 currentEnemy.IsBoss = true;
397                 currentEnemy.BonusDamage = 4;
398                 currentEnemy.Potions = 1;
399 
400                 AddLog("");
401                 AddHeaderLog("=== Final Boss ===");
402                 AddDefeatLog("An Ancient Dragon flies down from the sky
                      in a storm of fire and ash.");
403                 environmentLabel.Text = "Environment: a storm of fire and
                      ash";
404                 statusLabel.Text = "Final Boss battle. Choose your action
                      carefully.";
405             }
406 
407             UpdateUi();
408         }
409 
410         private void AttackButton_Click(object sender, EventArgs e)
411         {
412             TakeTurn(BattleAction.Attack);
413         }
414 
415         private void BlockButton_Click(object sender, EventArgs e)
416         {
417             TakeTurn(BattleAction.Block);
418         }
419 
```

```
420         private void HealButton_Click(object sender, EventArgs e)
421         {
422             TakeTurn(BattleAction.Heal);
423         }
424
425         private void TakeTurn(BattleAction playerAction)
426         {
427             if (player == null || currentEnemy == null || gameOver ||
                  waitingForUpgrade)
428             {
429                 return;
430             }
431
432             AddLog("");
433             AddHeaderLog(bossBattle ? "=== Boss Battle ===" : "=== Battle
                  ===");
434
435             BattleAction enemyAction = GetEnemyAction(currentEnemy,
                  bossBattle);
436
437             bool playerBlocked = playerAction == BattleAction.Block;
438             bool enemyBlocked = enemyAction == BattleAction.Block;
439
440             if (playerAction == BattleAction.Heal)
441             {
442                 if (player.Potions > 0)
443                 {
444                     int healAmount = player.UsePotion(random);
445                     AddPlayerLog("You used a potion and healed " +
                          healAmount + " health.");
446                 }
447                 else
448                 {
449                     AddWarningLog("You do not have any potions.");
450                 }
451             }
452             else if (playerAction == BattleAction.Attack)
453             {
454                 DoAttack(player, currentEnemy, enemyBlocked, true);
455             }
456             else
457             {
458                 AddInfoLog("You brace for impact and prepare to block.");
459             }
460
461             if (!currentEnemy.IsAlive())
462             {
463                 EndBattle(true);
464                 return;
```

```
465             }
466
467             if (enemyAction == BattleAction.Heal)
468             {
469                 int healAmount = currentEnemy.UsePotion(random);
470                 AddEnemyLog("The " + currentEnemy.Name + " used a potion
                      and healed " + healAmount + " health.");
471             }
472             else if (enemyAction == BattleAction.Attack)
473             {
474                 DoAttack(currentEnemy, player, playerBlocked, false);
475             }
476             else
477             {
478                 AddEnemyLog("The " + currentEnemy.Name + " prepares to
                      block.");
479             }
480
481             if (!player.IsAlive())
482             {
483                 EndBattle(false);
484                 return;
485             }
486
487             statusLabel.Text = "Choose your next action.";
488             UpdateUi();
489         }
490
491         private void DoAttack(Character attacker, Character defender,
              bool defenderBlocked, bool playerIsAttacking)
492         {
493             int damage = attacker.Attack(random);
494             bool critical = random.Next(1, 101) <=
                  attacker.GetCriticalChance();
495
496             if (critical)
497             {
498                 damage += random.Next(3, 7);
499             }
500
501             int originalDamage = damage;
502             bool countered = false;
503             int counterChance = 25;
504
505             if (defenderBlocked)
506             {
507                 damage = (int)Math.Ceiling(originalDamage * 0.25);
508
509                 if (damage < 1)
```

```
510                 {
511                     damage = 1;
512                 }
513 
514                 if (random.Next(1, 101) <= counterChance)
515                 {
516                     countered = true;
517                 }
518             }
519 
520             defender.Health -= damage;
521 
522             if (defender.Health < 0)
523             {
524                 defender.Health = 0;
525             }
526 
527             string attackerName;
528             string targetName;
529 
530             if (playerIsAttacking)
531             {
532                 attackerName = "Player";
533                 targetName = defender.Name;
534             }
535             else
536             {
537                 attackerName = "The " + attacker.Name;
538                 targetName = "player";
539             }
540 
541             if (defenderBlocked)
542             {
543                 if (critical)
544                 {
545                     if (playerIsAttacking)
546                     {
547                         AddPlayerLog(attackerName + " lands a critical hit with " + attacker.Weapon.Name +
548                                      " for " + originalDamage + " damage, but it was blocked down to " + damage + ".");
549                     }
550                     else
551                     {
552                         AddEnemyLog(attackerName + " lands a critical hit with " + attacker.Weapon.Name +
553                                     " for " + originalDamage + " damage, but it was blocked down to " +
```

```
                                        damage + ".");
554                     }
555                 }
556                 else
557                 {
558                     if (playerIsAttacking)
559                     {
560                         AddPlayerLog(attackerName + " attacks " +
                              targetName + " with " + attacker.Weapon.Name +
561                                      " for " + originalDamage + " damage,
                                        but it was blocked down to " +
                                        damage + ".");
562                     }
563                     else
564                     {
565                         AddEnemyLog(attackerName + " attacks " +
                              targetName + " with " + attacker.Weapon.Name +
566                                    " for " + originalDamage + " damage,
                                        but it was blocked down to " +
                                        damage + ".");
567                     }
568                 }
569             }
570             else
571             {
572                 if (critical)
573                 {
574                     if (playerIsAttacking)
575                     {
576                         AddPlayerLog(attackerName + " lands a critical
                              hit with " + attacker.Weapon.Name +
577                                      " and deals " + damage + " damage.");
578                     }
579                     else
580                     {
581                         AddEnemyLog(attackerName + " lands a critical hit
                              with " + attacker.Weapon.Name +
582                                    " and deals " + damage + " damage.");
583                     }
584                 }
585                 else
586                 {
587                     if (playerIsAttacking)
588                     {
589                         AddPlayerLog(attackerName + " attacks " +
                              targetName + " with " + attacker.Weapon.Name +
590                                      " and deals " + damage + " damage.");
591                     }
592                     else
```

```
593                     {
594                         AddEnemyLog(attackerName + " attacks " +
                              targetName + " with " + attacker.Weapon.Name +
595                                   " and deals " + damage + " damage.");
596                     }
597                 }
598             }
599 
600             if (countered && defender.IsAlive())
601             {
602                 attacker.Health -= originalDamage;
603 
604                 if (attacker.Health < 0)
605                 {
606                     attacker.Health = 0;
607                 }
608 
609                 string counterAttackerName;
610                 string counterTargetName;
611 
612                 if (playerIsAttacking)
613                 {
614                     counterAttackerName = "The " + defender.Name;
615                     counterTargetName = "you";
616                     AddWarningLog(counterAttackerName + " counters and
                          deals " + originalDamage + " damage to " +
                          counterTargetName + "!");
617                 }
618                 else
619                 {
620                     counterAttackerName = "You";
621                     counterTargetName = attacker.Name;
622                     AddVictoryLog(counterAttackerName + " counterattacks
                          and deals " + originalDamage + " damage to " +
                          counterTargetName + "!");
623                 }
624             }
625 
626             UpdateUi();
627         }
628 
629         private void EndBattle(bool playerWon)
630         {
631             SetBattleButtons(false);
632             UpdateUi();
633 
634             if (playerWon)
635             {
636                 if (bossBattle)
```

```
637                 {
638                     AddVictoryLog("The dragon crashes to the ground. You win!");
639                     AddVictoryLog("You defeated every enemy and slayed the Ancient Dragon!");
640                     statusLabel.Text = "Victory. Press Restart to begin a new run.";
641                     gameOver = true;
642                 }
643                 else
644                 {
645                     AddVictoryLog("You defeated the " + currentEnemy.Name + ".");
646 
647                     if (currentLevel <= 6)
648                     {
649                         waitingForUpgrade = true;
650                         SetUpgradeButtons(true);
651 
652                         if (currentLevel == 6)
653                         {
654                             statusLabel.Text = "Choose one final upgrade before the boss.";
655                             AddLog("");
656                             AddWarningLog("You survived the 6 levels. Choose one final upgrade before the Ancient Dragon.");
657                         }
658                         else
659                         {
660                             statusLabel.Text = "Choose one upgrade to continue.";
661                             AddLog("");
662                             AddWarningLog("Choose an upgrade.");
663                         }
664                     }
665                 }
666             }
667             else
668             {
669                 if (bossBattle)
670                 {
671                     AddDefeatLog("The dragon burns through your final defense.");
672                 }
673                 else
674                 {
675                     AddDefeatLog("The " + currentEnemy.Name + " defeated you.");
```

```
676                 }
677
678                 AddDefeatLog("Your journey ends in the Dark Ages.");
679                 statusLabel.Text = "Defeat. Press Restart to try again.";
680                 gameOver = true;
681                 SetUpgradeButtons(false);
682             }
683         }
684
685         private void UpgradeHealthButton_Click(object sender, EventArgs e)
686         {
687             if (!waitingForUpgrade || player == null)
688             {
689                 return;
690             }
691
692             player.MaxHealth += 10;
693             player.Health = player.MaxHealth;
694             AddInfoLog("Your max health is now " + player.MaxHealth +
                  ".");
695             FinishUpgrade();
696         }
697
698         private void UpgradeDamageButton_Click(object sender, EventArgs e)
699         {
700             if (!waitingForUpgrade || player == null)
701             {
702                 return;
703             }
704
705             player.BonusDamage += 2;
706             player.Health = player.MaxHealth;
707             AddInfoLog("Your bonus damage is now " + player.BonusDamage +
                  ".");
708             FinishUpgrade();
709         }
710
711         private void UpgradePotionButton_Click(object sender, EventArgs e)
712         {
713             if (!waitingForUpgrade || player == null)
714             {
715                 return;
716             }
717
718             player.Potions += 1;
719             player.Health = player.MaxHealth;
720             AddInfoLog("You now have " + player.Potions + " potion(s).");
721             FinishUpgrade();
722         }
```

```
723
724         private void FinishUpgrade()
725         {
726             waitingForUpgrade = false;
727             SetUpgradeButtons(false);
728             currentLevel++;
729             StartCurrentLevel();
730         }
731
732         private void SetBattleButtons(bool enabled)
733         {
734             attackButton.Enabled = enabled;
735             blockButton.Enabled = enabled;
736             healButton.Enabled = enabled;
737         }
738
739         private void SetUpgradeButtons(bool enabled)
740         {
741             upgradeHealthButton.Enabled = enabled;
742             upgradeDamageButton.Enabled = enabled;
743             upgradePotionButton.Enabled = enabled;
744         }
745
746         private void UpdateUi()
747         {
748             if (player == null)
749             {
750                 return;
751             }
752
753             levelLabel.Text = bossBattle ? "Level: Final Boss" : "
                  Level: " + currentLevel;
754             classLabel.Text = "Class: " + player.ClassName;
755             playerHealthLabel.Text = "Health: " + player.Health + "/" +
                  player.MaxHealth;
756             weaponLabel.Text = "Weapon: " + player.Weapon.Name;
757             bonusDamageLabel.Text = "Bonus Damage: " + player.BonusDamage;
758             potionLabel.Text = "Potions: " + player.Potions;
759
760             playerHealthBar.Maximum = Math.Max(1, player.MaxHealth);
761             playerHealthBar.Value = Math.Max(0, Math.Min(player.Health,
                  playerHealthBar.Maximum));
762
763             if (currentEnemy != null)
764             {
765                 enemyNameLabel.Text = "Enemy: " + currentEnemy.Name + " "
                      + currentEnemy.EnemyClassName + " (" +
                      currentEnemy.Weapon.Name + ")";
766                 enemyHealthLabel.Text = "Enemy Health: " +
```

```
                    currentEnemy.Health + "/" + currentEnemy.MaxHealth;

                enemyHealthBar.Maximum = Math.Max(1,
                  currentEnemy.MaxHealth);
                enemyHealthBar.Value = Math.Max(0, Math.Min
                  (currentEnemy.Health, enemyHealthBar.Maximum));
            }
            else
            {
                enemyNameLabel.Text = "Enemy: -";
                enemyHealthLabel.Text = "Enemy Health: -";
                enemyHealthBar.Value = 0;
            }
        }

        private void AddLog(string text)
        {
            AddLog(text, Color.WhiteSmoke);
        }

        private void AddLog(string text, Color color)
        {
            logBox.SelectionStart = logBox.TextLength;
            logBox.SelectionLength = 0;
            logBox.SelectionColor = color;
            logBox.AppendText(text + Environment.NewLine);
            logBox.SelectionColor = logBox.ForeColor;
            logBox.ScrollToCaret();
        }

        private void AddHeaderLog(string text)
        {
            AddLog(text, Color.Gold);
        }

        private void AddPlayerLog(string text)
        {
            AddLog(text, Color.LightGreen);
        }

        private void AddEnemyLog(string text)
        {
            AddLog(text, Color.Salmon);
        }

        private void AddInfoLog(string text)
        {
            AddLog(text, Color.LightSkyBlue);
        }
```

```
813
814         private void AddWarningLog(string text)
815         {
816             AddLog(text, Color.Khaki);
817         }
818
819         private void AddVictoryLog(string text)
820         {
821             AddLog(text, Color.MediumSpringGreen);
822         }
823
824         private void AddDefeatLog(string text)
825         {
826             AddLog(text, Color.OrangeRed);
827         }
828
829         private BattleAction GetEnemyAction(Enemy enemy, bool isBoss)
830         {
831             if (enemy.Potions > 0 && enemy.Health <= enemy.MaxHealth / 3)
832             {
833                 int healChance = random.Next(1, 101);
834
835                 if (healChance <= 40)
836                 {
837                     return BattleAction.Heal;
838                 }
839             }
840
841             int blockChance;
842
843             if (isBoss)
844             {
845                 blockChance = 18;
846             }
847             else
848             {
849                 blockChance = 24;
850             }
851
852             if (enemy.Weapon.Type == WeaponType.Bow || enemy.Weapon.Type
                  == WeaponType.Wand)
853             {
854                 blockChance -= 8;
855             }
856
857             if (enemy.Health < enemy.MaxHealth / 2)
858             {
859                 blockChance += 8;
860             }
```

```
861
862             int roll = random.Next(1, 101);
863
864             if (roll <= blockChance)
865             {
866                 return BattleAction.Block;
867             }
868
869             return BattleAction.Attack;
870         }
871
872         private Enemy CreateEnemy(int level)
873         {
874             int classNumber = random.Next(1, 6);
875             string className = "";
876             Weapon weapon = null;
877
878             if (classNumber == 1)
879             {
880                 className = "Knight";
881                 weapon = new Weapon(WeaponType.Sword, "Sword", 5, 9);
882             }
883             else if (classNumber == 2)
884             {
885                 className = "Vanguard";
886                 weapon = new Weapon(WeaponType.BattleAxe, "Battle Axe",
                      6, 10);
887             }
888             else if (classNumber == 3)
889             {
890                 className = "Hunter";
891                 weapon = new Weapon(WeaponType.Bow, "Bow", 4, 10);
892             }
893             else if (classNumber == 4)
894             {
895                 className = "Rogue";
896                 weapon = new Weapon(WeaponType.Dagger, "Daggers", 4, 9);
897             }
898             else
899             {
900                 className = "Mage";
901                 weapon = new Weapon(WeaponType.Wand, "Wand", 5, 10);
902             }
903
904             string enemyName = enemyNames[random.Next(enemyNames.Length)];
905
906             while (enemyName.ToLower().Contains(className.ToLower()) ||
                  className.ToLower().Contains(enemyName.ToLower()))
907             {
```

```
908                     enemyName = enemyNames[random.Next(enemyNames.Length)];
909                 }
910 
911                 int health = 22 + (level * 4) + random.Next(0, 5);
912                 Enemy enemy = new Enemy(enemyName, className, health, weapon);
913 
914                 if (level >= 4)
915                 {
916                     enemy.BonusDamage = 1;
917                 }
918 
919                 if (level >= 6)
920                 {
921                     enemy.Potions = 1;
922                 }
923 
924                 return enemy;
925             }
926         }
927 }
```